

Modeling Trajectories From Time-Series Outcomes in Clinical Trials

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Disclosures

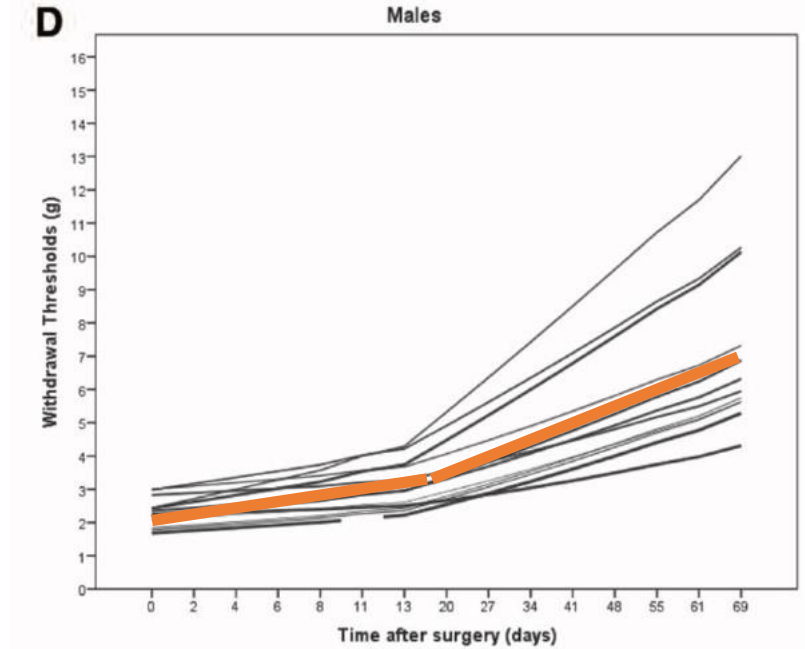
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- American Society of Anesthesiologists
 - Statistical Editor, *Anesthesiology*
- American Headache Society
 - Statistical Editor, *Headache*

Objectives

- Quick tour of 'Trajectory Analysis' as applied in clinical trials
- Overview of benefits and weaknesses of the approach
- Examples of a few applications
 - Pain
 - Physical activity

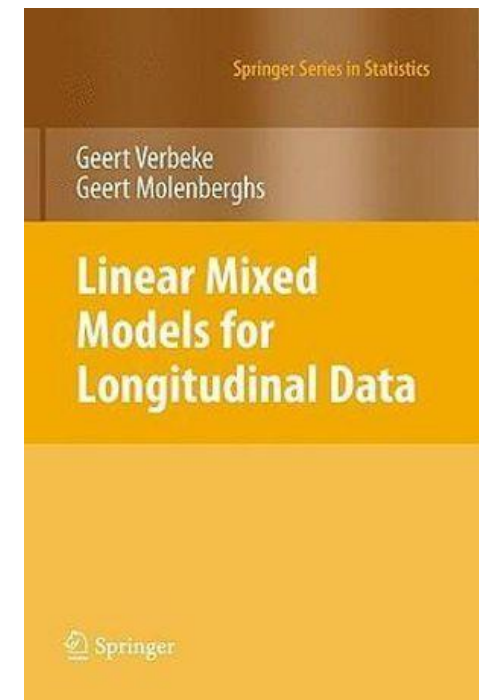
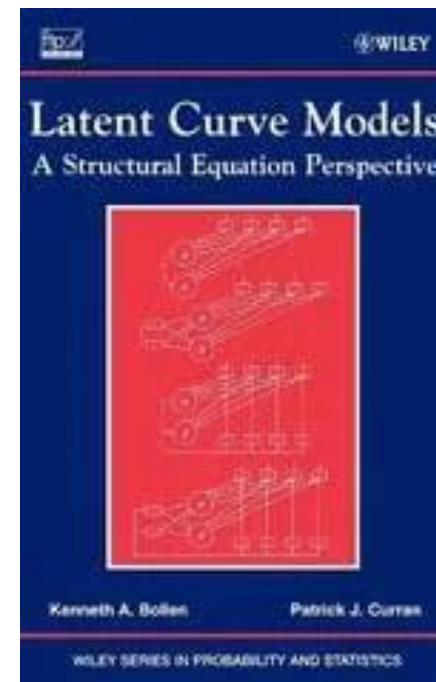
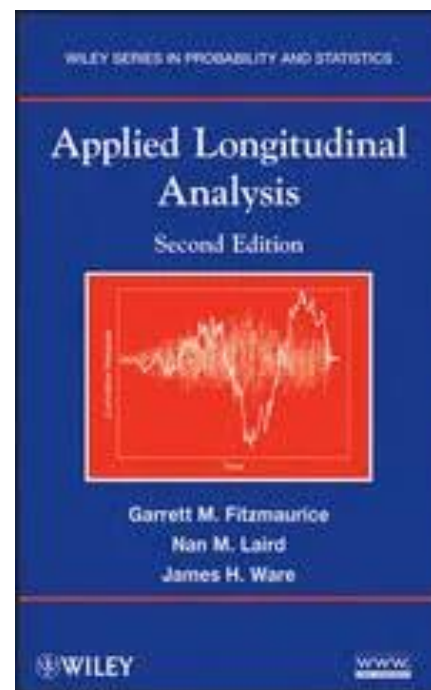
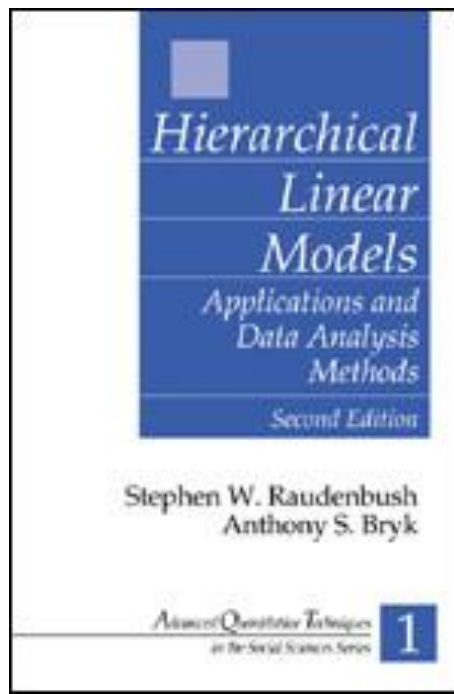
Trajectory Analysis

- (Many) Repeated Measures
- Study of individual change
 - Stability of change
 - Timing/nature of change
 - Heterogeneity across individuals
- Group-level estimates from individual trajectories
 - Can have missing data
 - Can have different measurement occasions



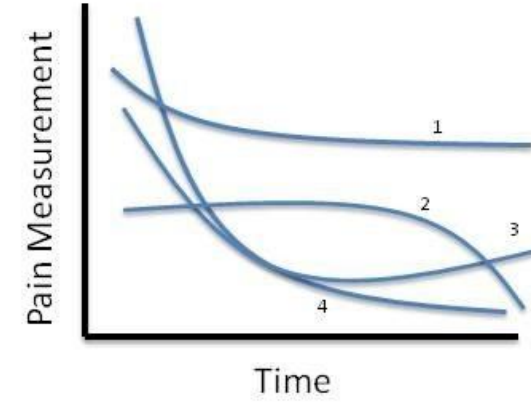
Similar Approaches

- Linear-mixed models for longitudinal data
- Multilevel models
- (Latent) Growth curve analysis
- Multilevel polynomial regression
- Hierarchical linear models (HLM)
- Trajectory analysis



Level 1

$$Pain_i = \pi_{0j} + \pi_{1j}(time) + \pi_{2j}(time^2) + r_{ij}, \text{ where } r_{ij} \sim N(0, \sigma^2)$$



Level 1

Using random effects, a unique pain trajectory is specified for each individual in the study

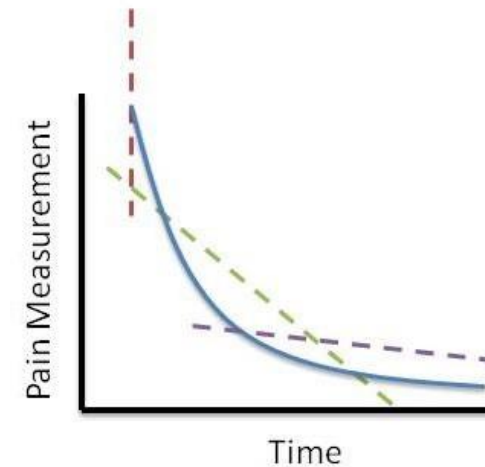
Four individuals are displayed (right) with different pain trajectories

Level 2

$$\pi_{0j} = \beta_{00} + \beta_{01}(\text{Predictor1}) + \beta_{02}(\text{Predictor2})... + u_{0j},$$

$$\pi_{1j} = \beta_{10} + \beta_{11}(\text{Predictor1}) + \beta_{12}(\text{Predictor2})... + u_{1j}, \text{ where } \begin{pmatrix} u_{0j} \\ u_{1j} \\ u_{2j} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_{00} & \tau_{01} & \tau_{02} \\ \tau_{10} & \tau_{11} & \tau_{12} \\ \tau_{20} & \tau_{21} & \tau_{22} \end{pmatrix} \right]$$

$$\pi_{2j} = \beta_{20} + \beta_{21}(\text{Predictor2}) + \beta_{22}(\text{Predictor2})... + u_{2j},$$



Level 2

Each parameter in Level 1 is treated as an outcome in level 2

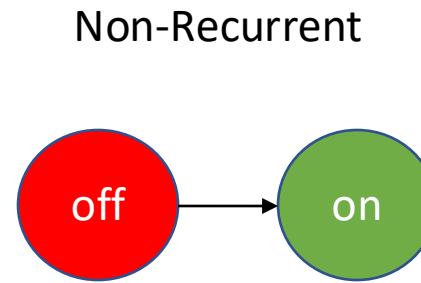
- π_{0j} Variability in starting values across individuals
- π_{1j} Variability in slope across individuals
- π_{2j} Variability in deceleration across individuals

Outcome Consideration

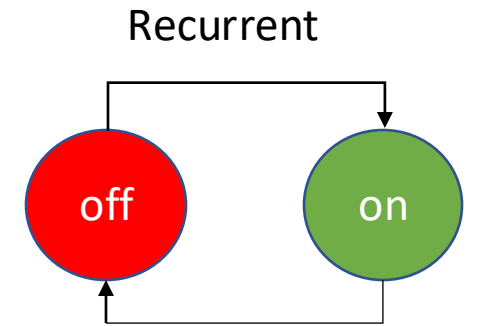
- Nominal Events
 - Binary
 - Example: Stroke, Death
- Ordinal Scaling
 - Example: COVID symptoms
- Continuous scaling
 - Example: pain, weight, physical activity, self-reported disability

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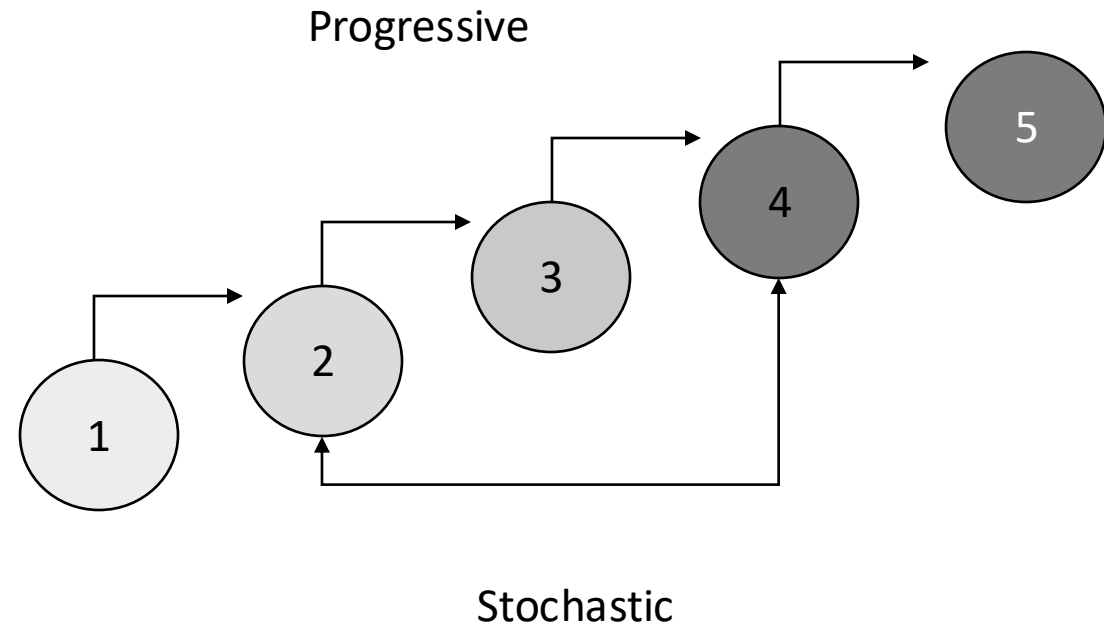
Time-to-event
Survival
Cox Model
Kaplan-Meier



Recurrent time-to-event
Marginal rates
Frailty models
Markov states

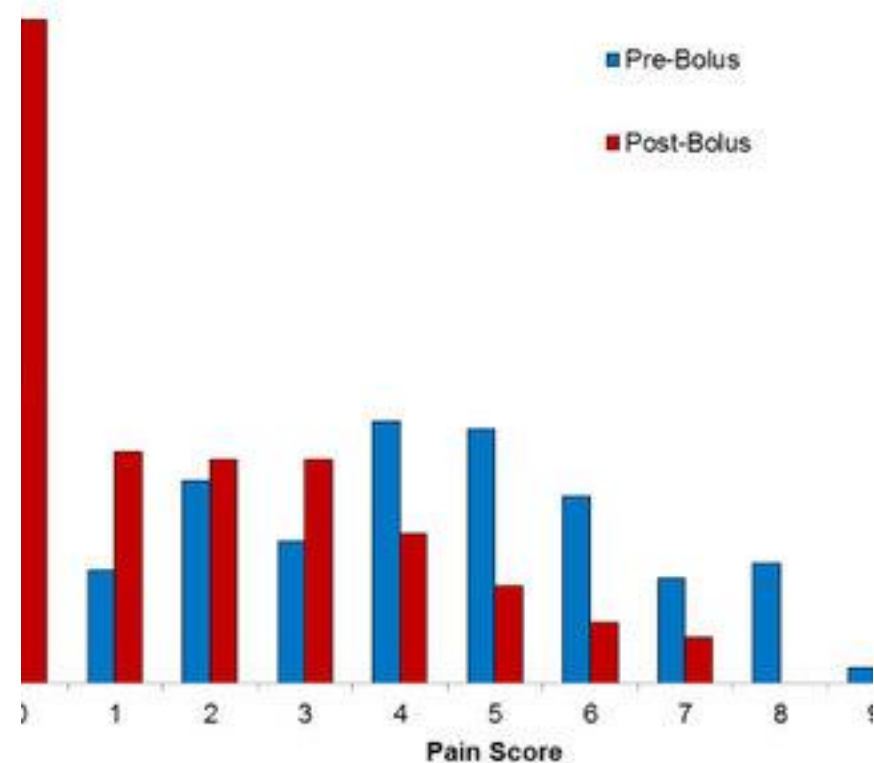
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Research article | [Open Access](#) | [Published: 27 October 2003](#)

How many repeated measures in repeated measures designs? Statistical issues for comparative trials

[Andrew J Vickers](#) 

[BMC Medical Research Methodology](#) **3**, Article number: 22 (2003)

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Modeling Individual Recovery after Peripheral Nerve Injury in Rats and the Effects of Parturition

Carol A. Aschenbrenner, M.A., Timothy T. Houle, Ph.D., Silvia Gutierrez, Ph.D., James C. Eisenach, M.D.

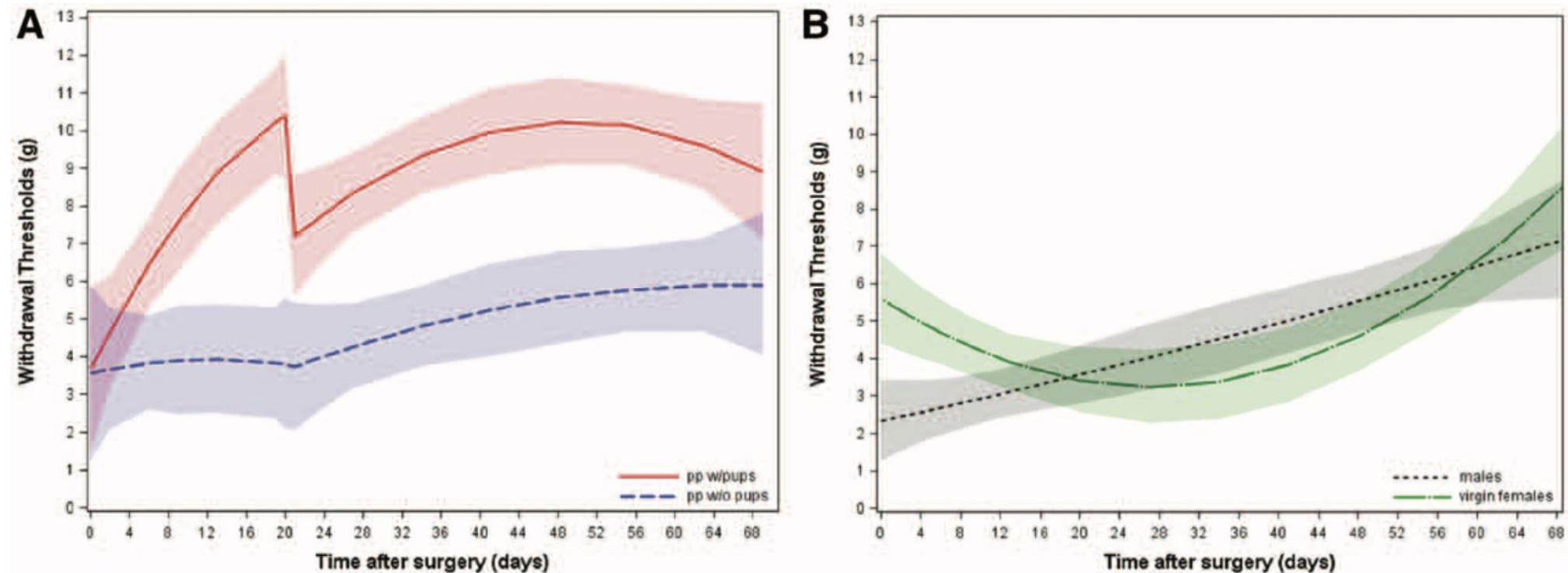


Fig. 2. Postpartum and nonpostpartum group trajectories. *Lines* represent predicted values and shaded areas represent 95% CIs of modeled trajectories. (A) Modeled trajectory when partial spinal nerve ligation is performed on day 0 in postpartum-with-pups (pp w/pups; red) or postpartum-separated-from pups (pp w/o pups; blue) animals. (B) Modeled trajectory when partial spinal nerve ligation is performed on day 0 in virgin females (green) or male (gray) rats.

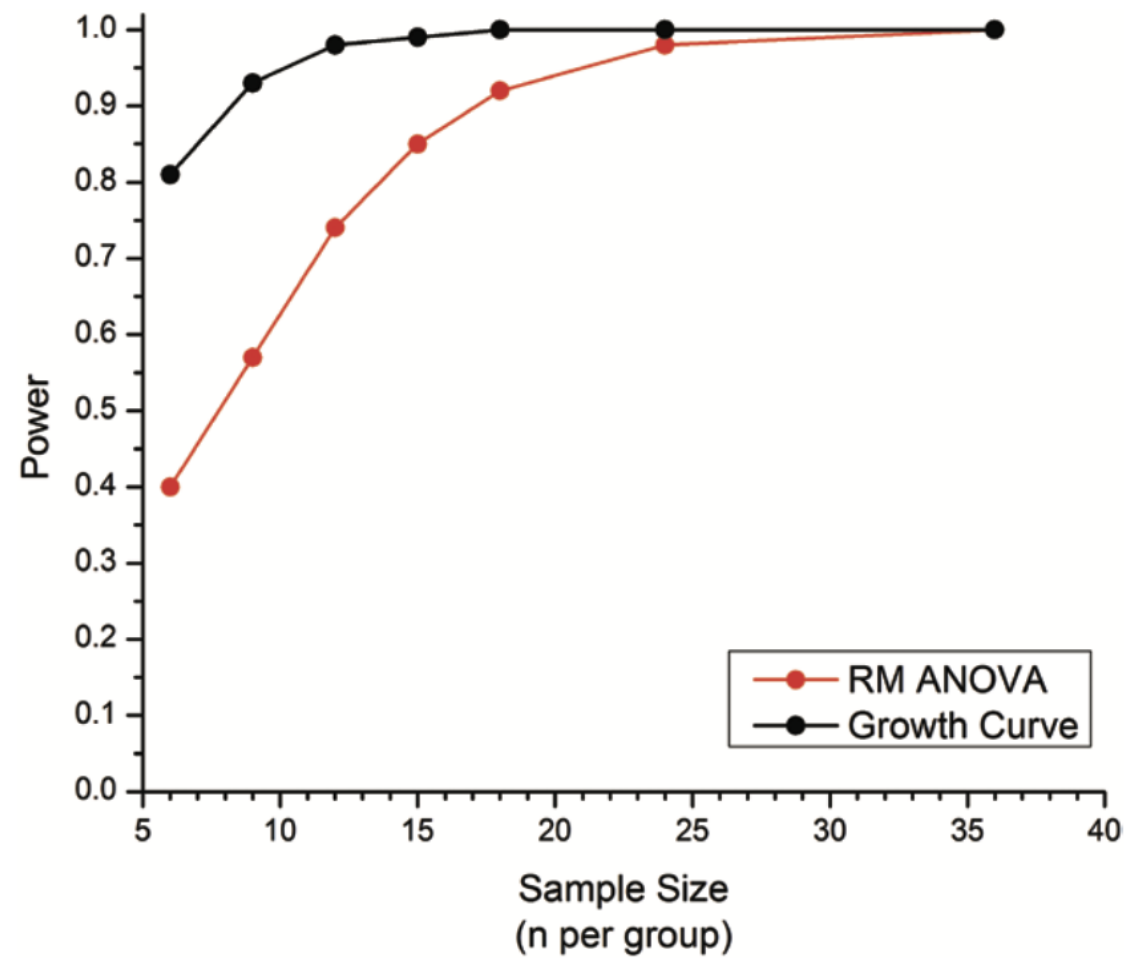


Fig. 6. Plot of statistical power (y-axis) of repeated measures (RM) ANOVA model (*red*) and growth curve model (*black*) obtained from analyses of simulated data by sample size (n per group) (x-axis). The two simulations estimated two different parameters from the approaches and are not different estimates of the same construct. In the ANOVA simulations, the group $\times$ time interaction effect was examined, whereas a model fit was the focus of the growth curve simulations.

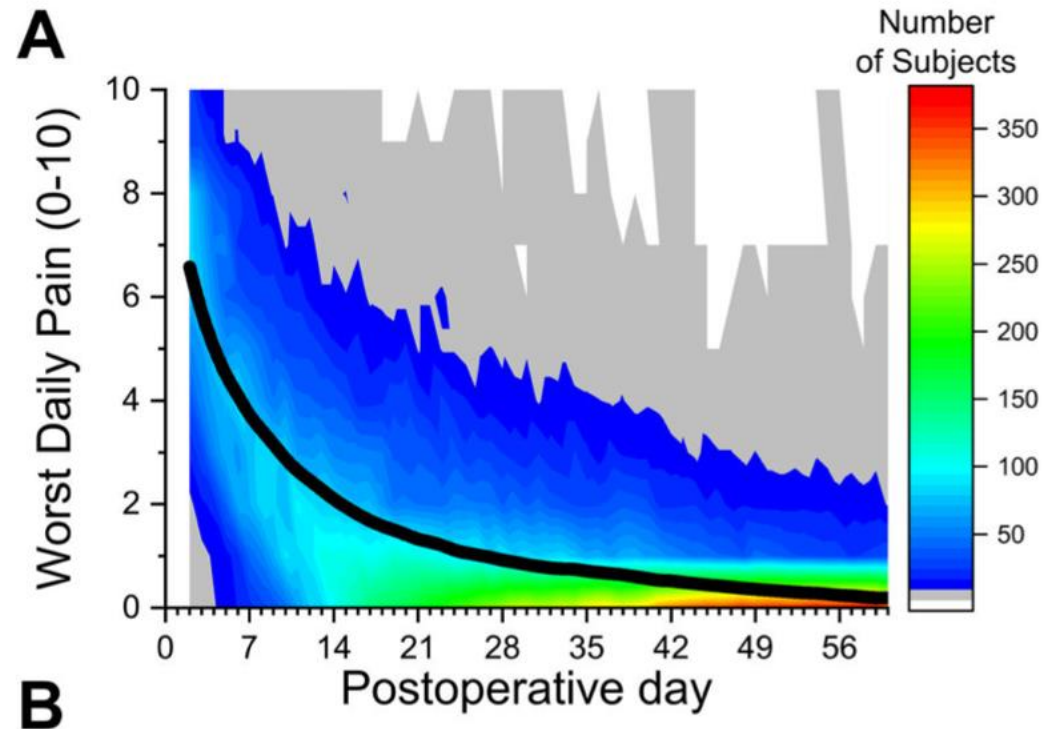
Patterns of recovery from pain after cesarean delivery

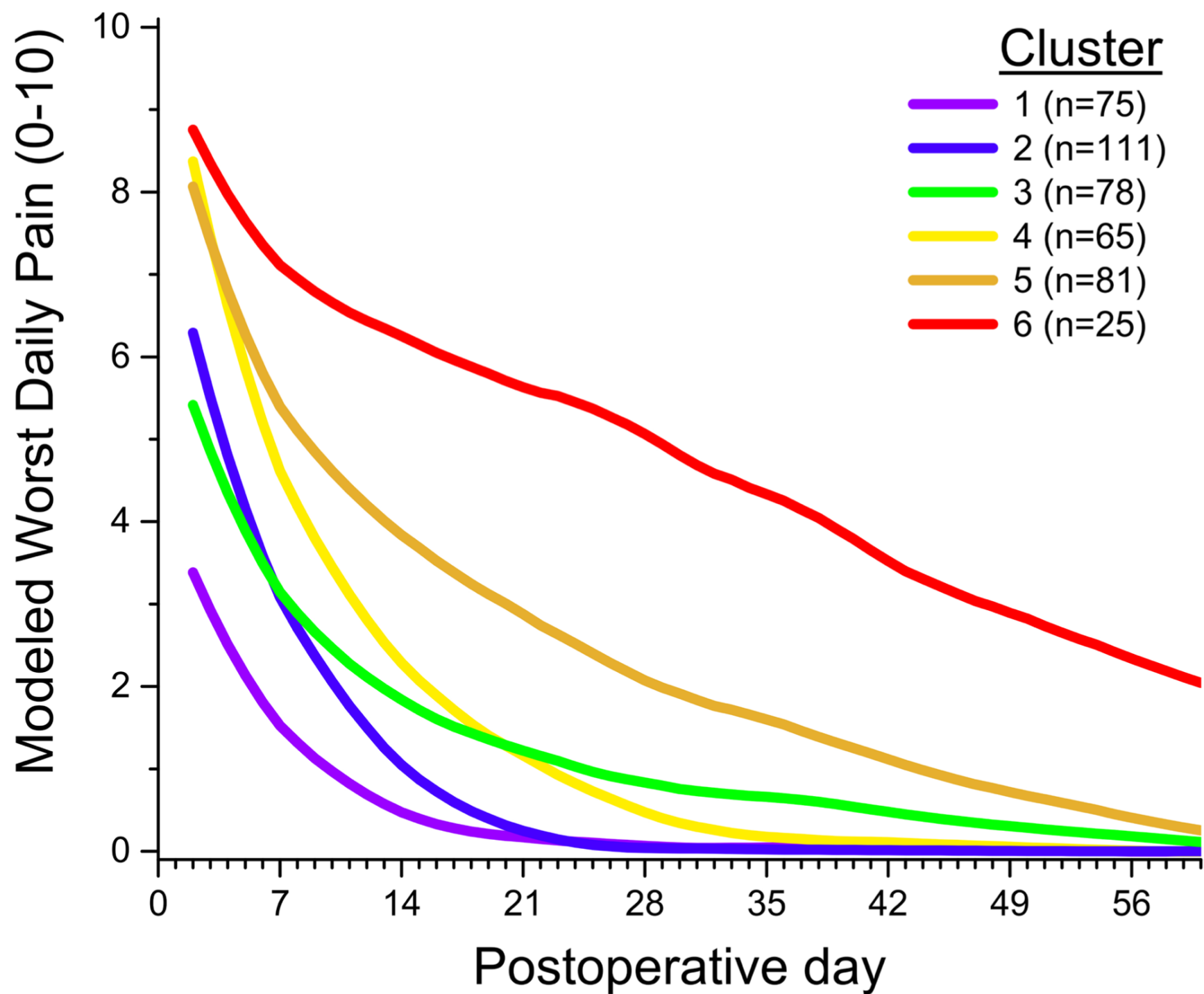
Jessica L. Booth^a, Emily E. Sharpe^b, Timothy T. Houle^c, Lynnette Harris^a, Regina S. Curry^a, Carol A. Aschenbrenner^a, and James C. Eisenach^a

^aDepartment of Anesthesiology, Wake Forest School of Medicine, Winston-Salem, NC;

^bDepartment of Anesthesiology, Mayo Clinic College of Medicine, Rochester, MN;

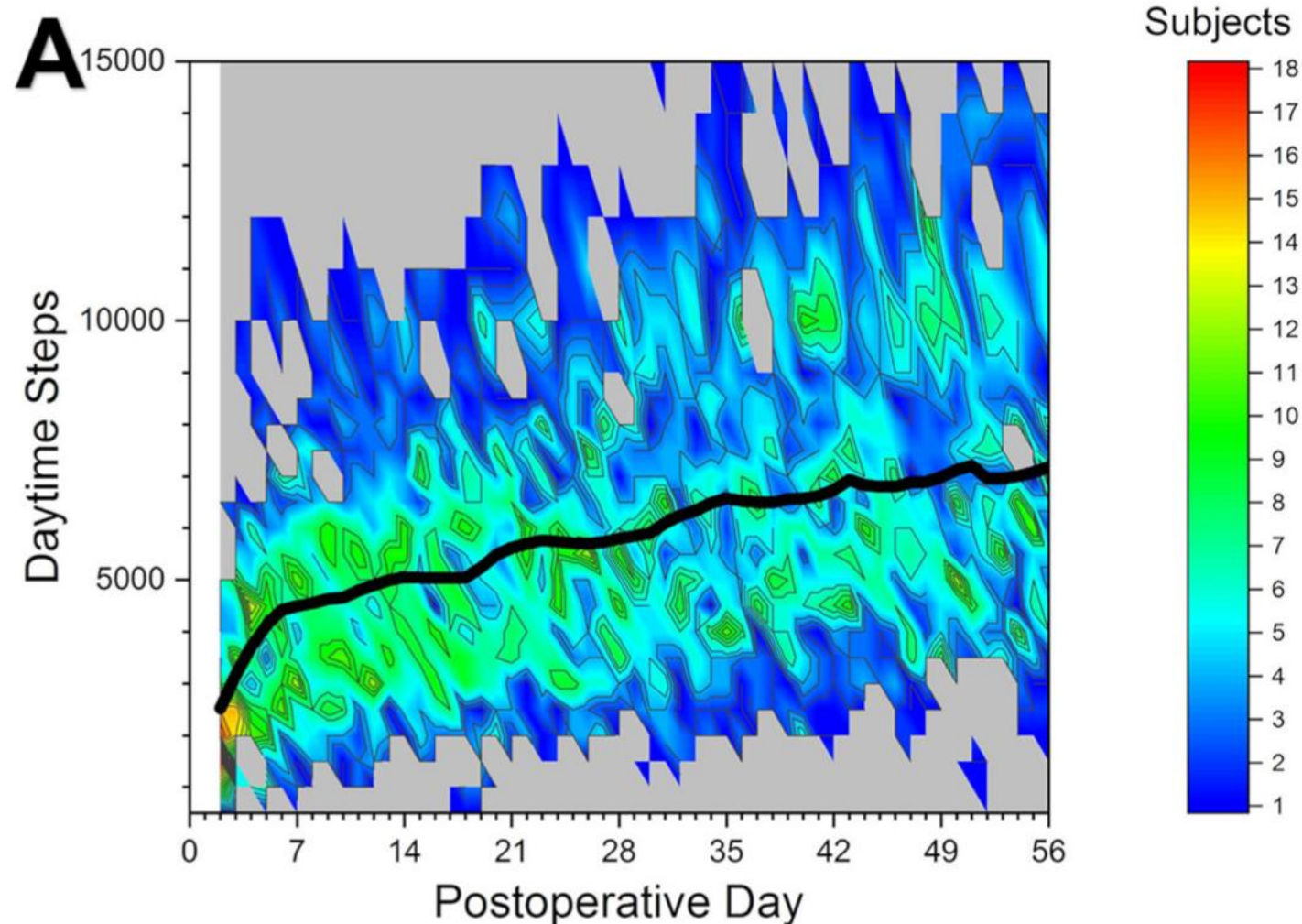
^cDepartment of Anesthesiology and Perioperative Medicine, Massachusetts General Hospital, Harvard Medical School, Boston, MA.

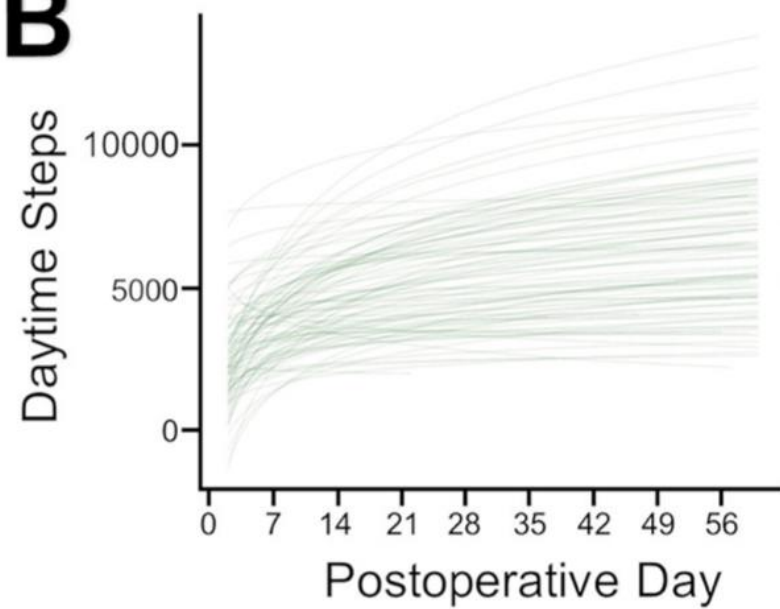
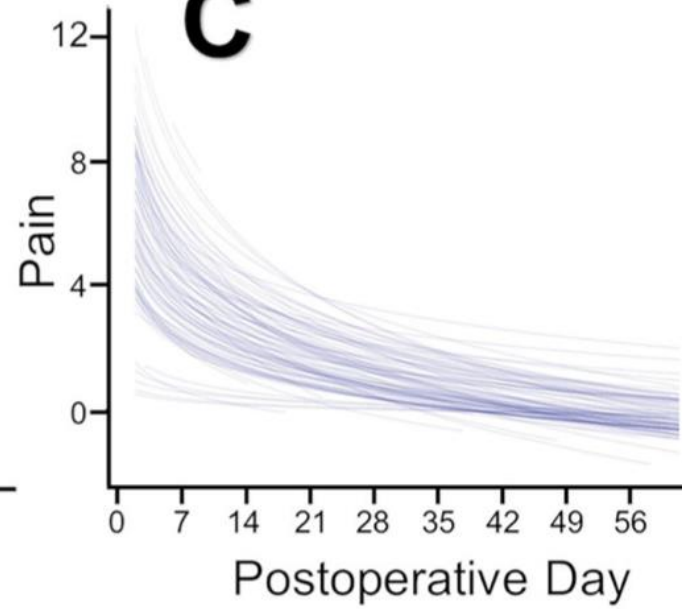
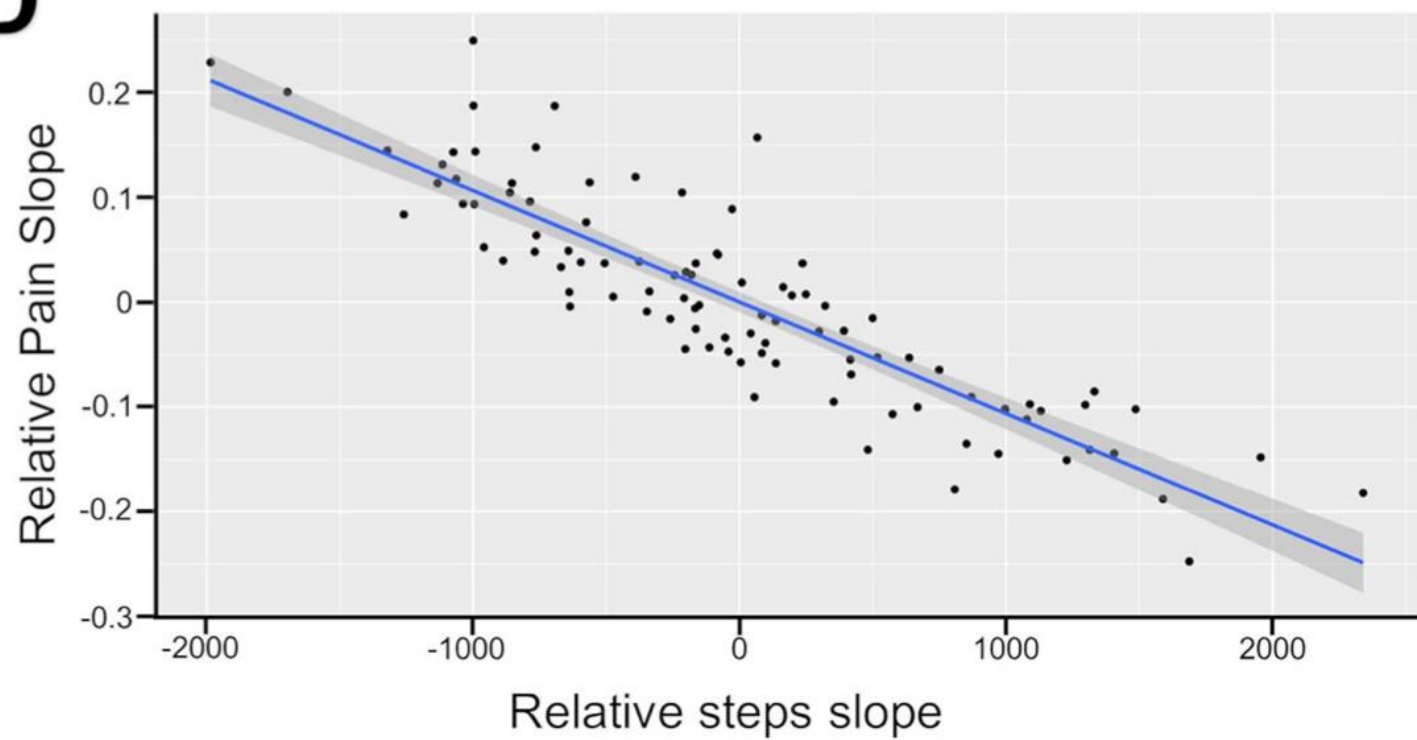




Recovery of physical activity after cesarean delivery and its relationship with pain

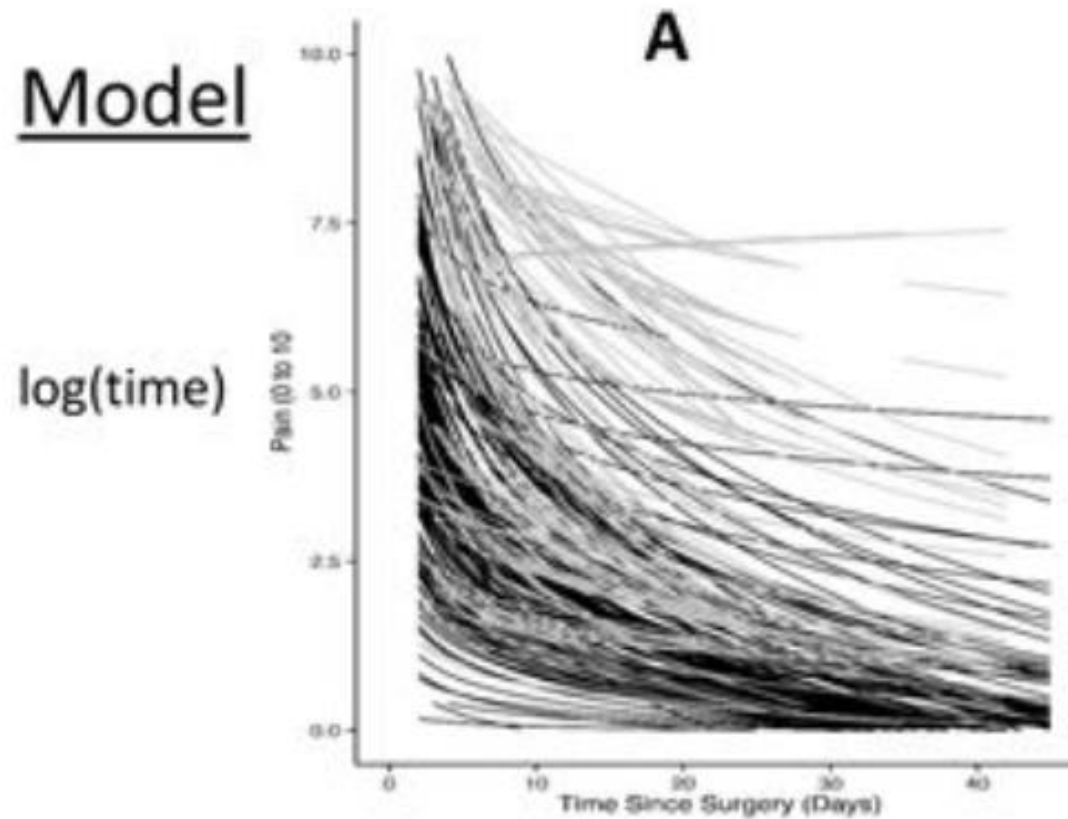
Emily E. Sharpe¹, Jessica L. Booth², Timothy T. Houle³, Peter H. Pan⁴, Lynnette C. Harris⁴, Carol A. Aschenbrenner⁴, James C. Eisenach⁴



B**C****D**

Day-to-day experience in resolution of pain after surgery

Timothy T. Houle^a, Scott Miller^b, Jason E. Lang^c, Jessica L. Booth^b, Regina S. Curry^b, Lynnette Harris^b, Carol A. Aschenbrenner^b, and James C. Eisenach^b



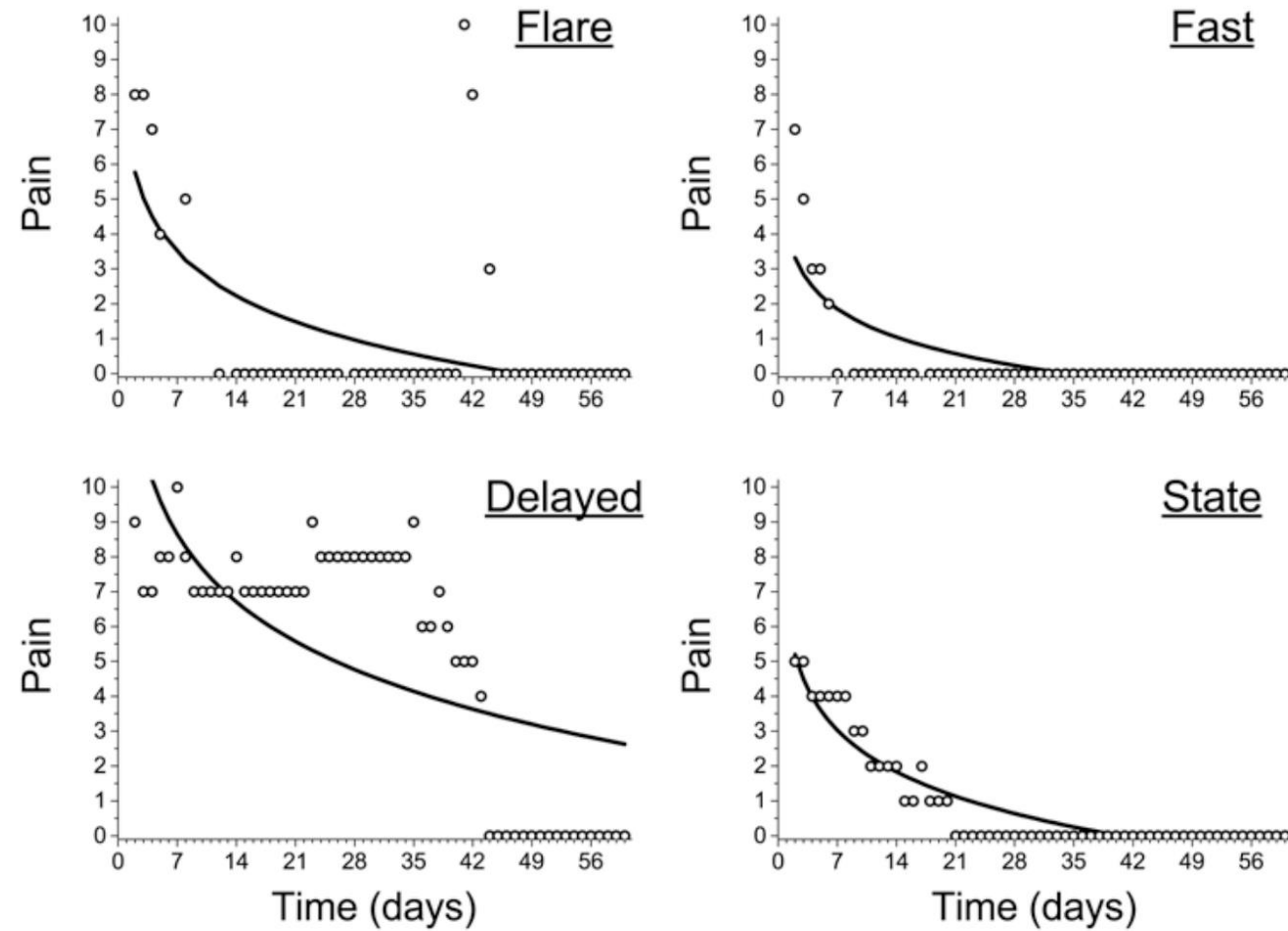


Figure 4.

Examples of change processes that induce a pattern of errors into the residuals. In each example, the distance from the prediction line to the observed point contains a temporal pattern such that points nearby can be used to predict the error of subsequent points. This can be the case because of many “0” pain scores in a sequence (“fast”), because a small group of points are very aberrant (“flare”), because the change process is not typical of a log(time model) (delayed), or because the pain scores follow a state-like pattern where the only change levels occasionally.

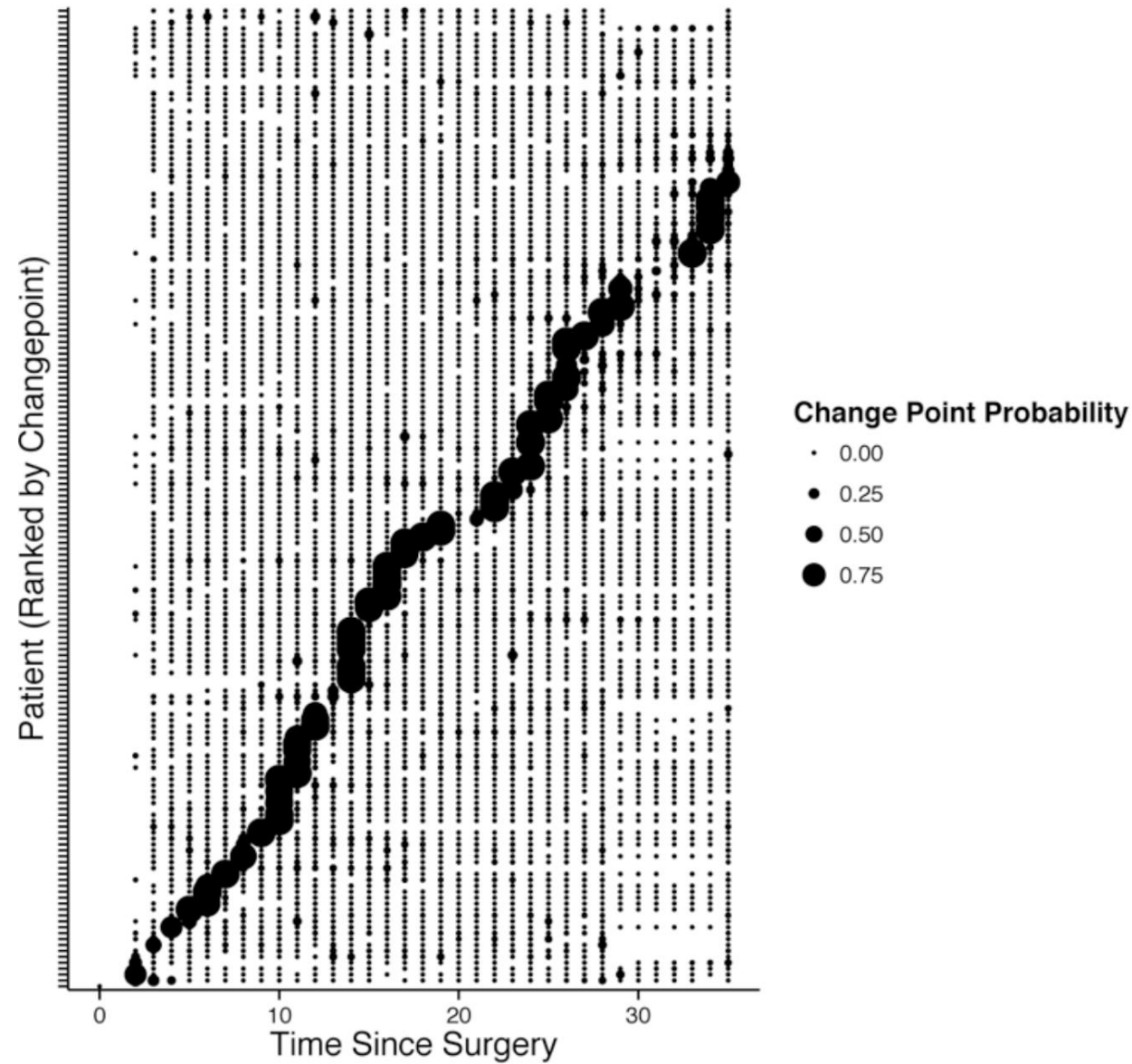


Figure 5.
The location of the change-point (x-axis) for each individual participant (each row on the y-axis). The size of the circle reflects the probability that there is a change point at that time for that participant.

Thank You!

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